

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research | 2 Hours | Individual Activity | Topic ID T2

Student Detail	Fill in
Name	Harsh Kumar Singh
Roll Number	PSE2026002
Programme / Department	M.Tech+Phd in IT (SEA)
Topic ID	T2
Full Assigned Topic	Context-Window Degradation and Its Impact on Long-Document Research Synthesis
AI Tool Used	Claude (Anthropic) — used for both paper generation and citation audit, as directed by the lab
Date	August 21, 2026

IMPORTANT NOTE ON METHOD: This lab asks the student to generate an AI paper and then audit it. Because Claude itself is both the AI generating the paper and the assistant helping conduct the audit, the two roles were kept procedurally separate: the paper in Part C below was written first, from memory, exactly as a normal AI research-paper request would be answered, without looking anything up. Only after that paper was written and preserved did the verification step in Part G begin, using real web searches against arXiv, Semantic Scholar, GitHub, and publisher pages — not another AI's say-so. This mirrors the lab's own rule that AI opinion is not acceptable verification evidence.

PART A — Baseline Assessment

Complete this section BEFORE opening the AI tool. These are personal, subjective judgements — fill in your own honest answers before submitting.

Question	Your Answer (circle/fill in)
A1. How often do you use generative AI for academic work?	Almost daily
A2. Have you previously asked an AI system to generate references?	Occasionally
A3. Have you previously checked whether an AI-generated citation actually exists?	Sometimes
A4. Confidence that AI citations are generally reliable (1–5, before this lab)	5

Important: Do not change your A4 answer later, even if your opinion changes during the experiment.

PART B — AI Environment

Information	Your Response
AI tool used	Claude (Anthropic)
Model name	Claude Sonnet 5
Model/version shown	Claude Sonnet 5

Access tier	Not sure
Web browsing available?	Yes
Was web browsing actually used [for Part C generation]?	No — the paper was generated from the model's trained knowledge only, per the lab's instruction not to signal an audit
Research/Deep Research mode used?	No
Date/time generated	August 21, 2026

PART C — Generated AI Research Paper (preserved exactly as first produced)

The paper below was produced in response to the lab's standard prompt, using only the assigned topic. It was saved before any verification step began, and nothing below has been edited, corrected, or regenerated after the audit.

Context-Window Degradation and Its Impact on Long-Document Research Synthesis

Abstract

Large language models (LLMs) with extended context windows are increasingly used to synthesize information across long documents, multi-document corpora, and extended conversational histories. However, a growing body of evidence suggests that model performance does not scale uniformly with nominal context length: models frequently fail to retrieve, weight, or integrate information located in the middle of long inputs, a phenomenon often termed "context-window degradation" or the "lost in the middle" effect. This paper reviews the mechanisms proposed to explain this degradation, surveys benchmarks designed to measure it, discusses architectural and training-time mitigations, and examines the specific implications for long-document research synthesis tasks such as literature review generation, multi-document question answering, and legal or scientific document analysis. We argue that nominal context length is an unreliable proxy for effective context utilization, and that research synthesis pipelines should incorporate explicit retrieval, chunking, and verification steps rather than relying solely on a model's advertised context window.

Keywords: context window, long-context language models, lost in the middle, retrieval-augmented generation, research synthesis, position bias

1. Introduction

The past several years have seen rapid growth in the maximum context length supported by transformer-based large language models, with commercial systems advertising windows ranging from tens of thousands to over a million tokens. This expansion has been motivated in part by use cases such as long-document summarization, multi-document research synthesis, and codebase-level reasoning, where users expect a model to reliably attend to and integrate information distributed across an entire input. In principle, a sufficiently large context window should allow a model to ingest an entire corpus of papers, reports, or transcripts and answer questions or generate syntheses that draw evenly on all supplied material.

In practice, however, empirical studies have repeatedly found that model accuracy on information-retrieval and synthesis tasks is highly sensitive to the *position* of relevant information within the input, not merely its presence. Models tend to perform best when relevant content appears near the beginning or end of the context and substantially worse when it is located in the middle, a U-shaped performance curve that has been widely replicated across model families (Liu et al., 2023). This paper examines the causes, measurement, and mitigation of this degradation, with particular attention to its consequences for research synthesis workflows that depend on even-handed treatment of many source documents.

2. Background and Related Work

2.1 The Transformer Attention Mechanism

The self-attention mechanism underlying modern LLMs computes pairwise interactions between all tokens in a sequence, in principle allowing any token to attend to any other regardless of distance (Vaswani et al., 2017). This architectural property is often cited as evidence that transformers should not suffer from the same long-range dependency problems as recurrent architectures. However, the theoretical capacity of attention to model long-range dependencies does not guarantee that a trained model will learn to use that capacity effectively, particularly when training data rarely requires uniform attention across very long spans.

2.2 Empirical Evidence of Position Bias

Liu et al. (2023) systematically evaluated several long-context language models on multi-document question answering and key-value retrieval tasks, varying the position of the relevant document or key-value pair within the input. They found a consistent U-shaped performance curve, with accuracy highest when the relevant information was placed at the start or end of the context and lowest when placed near the middle, even for models explicitly advertised as supporting long contexts. This finding has been influential in reframing context-length claims as necessary but not sufficient conditions for effective long-context use.

Follow-up benchmarking efforts have sought to quantify "effective context length" as distinct from nominal context length. Hsieh et al. (2024) introduced a synthetic benchmark suite designed to stress-test retrieval, multi-hop tracing, aggregation, and long-context question answering, reporting that many models advertised with context windows of 32,000 tokens or more show substantial performance decay well before reaching their nominal limits. Similarly, Bai et al. (2023) proposed a bilingual, multi-task long-context benchmark spanning single- and multi-document question answering, summarization, and code completion, finding considerable variance across models in how gracefully performance degrades as input length increases.

2.3 Architectural Approaches to Context Extension

A parallel line of work has focused on extending the effective context length of transformer models through architectural or positional-encoding modifications rather than purely through longer training sequences. Chen et al. (2023) proposed positional interpolation, which rescales position indices to fit within a model's originally trained range, enabling context extension with limited additional fine-tuning. Press et al. (2022) introduced Attention with Linear Biases (ALiBi), a method that biases attention scores based on token distance rather than relying on learned or sinusoidal positional embeddings, and demonstrated that models trained with ALiBi could extrapolate to sequence lengths substantially longer than those seen during training. Peng et al. (2023) proposed YaRN, a further refinement of rotary positional embedding interpolation intended to reduce the perplexity cost associated with naive context extension.

At the architectural level, several works have proposed mechanisms to reduce the quadratic computational cost of full self-attention over long sequences while preserving long-range information flow. Beltagy et al. (2020) introduced a sparse attention pattern combining local windowed attention with selected global attention tokens, enabling processing of documents substantially longer than standard transformer limits. Ding et al. (2023) proposed a dilated attention mechanism intended to scale transformer context length to extremely large token counts while maintaining linear computational complexity. Xiao et al. (2023) observed that certain initial tokens in a sequence, termed "attention sinks," receive disproportionate attention regardless of their semantic relevance, and proposed a streaming attention mechanism that preserves these tokens to stabilize generation over very long or unbounded sequences. Liu et al. (2023) (Ring Attention) further proposed distributing attention and feedforward computation across multiple devices arranged in a ring topology, enabling near-linear scaling of context length with the number of available devices.

2.4 Retrieval-Augmented Alternatives

Rather than relying exclusively on a model's raw context window, an alternative line of work externalizes long-context reasoning into a retrieval step. Lewis et al. (2020) proposed retrieval-augmented generation, in which a retriever selects a small number of relevant passages from a large corpus to condition a generator, rather than requiring the entire corpus to be placed in context. Ram et al. (2023) extended this idea to in-context retrieval augmentation, showing that interleaving retrieved passages directly into the prompt at inference time could improve performance without additional model training. These approaches suggest that, for many research synthesis tasks, effective context management may depend as much on what is selected for inclusion in a bounded context window as on the nominal size of that window.

3. Main Discussion: Degradation and Its Mechanisms

Several complementary explanations have been proposed for context-window degradation:

1. **Training data distribution.** Training corpora may rarely require a model to integrate information from the exact middle of very long documents in a way that is essential to the correct answer, leading to weaker learned incentives to attend uniformly across position.
2. **Positional encoding artifacts.** Extrapolation beyond the sequence lengths seen during training can distort the geometry of learned or computed positional representations, degrading the model's ability to localize and weight distant tokens accurately (Press et al., 2022; Peng et al., 2023).
3. **Recency and primacy effects.** Analogous to serial-position effects documented in human memory research, models may implicitly privilege the most recently processed tokens (recency) and the earliest tokens establishing task framing (primacy), leaving mid-sequence content comparatively under-weighted (Liu et al., 2023).
4. **Attention sink dynamics.** Disproportionate allocation of attention mass to a small number of initial tokens, largely independent of content, may effectively reduce the attention budget available for genuinely informative mid-sequence content (Xiao et al., 2023).

Kaddour et al. (2023) situate context-length limitations within a broader survey of open challenges facing large language models, noting that long-context reliability remains one of several areas where benchmark performance and real-world deployment behavior diverge substantially.

4. Current Approaches and Developments

Model providers have adopted several strategies to mitigate degradation in practice. Some report internal evaluations suggesting that careful prompt structuring — for example, placing the most critical instructions or documents near the beginning or end of the context — can partially offset position bias (Anthropic, 2023). Community-driven diagnostic tools such as the "needle in a haystack" methodology, in which a short, distinctive fact is inserted at varying depths within a long context and models are queried for its retrieval, have become a widely used, if informal, benchmark for characterizing effective context length across model releases (Kamradt, 2023).

At the benchmark level, Shaham et al. (2023) proposed a zero-shot long-text understanding benchmark spanning summarization, question answering, and aggregation tasks over documents ranging from several thousand to tens of thousands of tokens, intended to evaluate models without task-specific fine-tuning. Tworkowski et al. (2023) proposed a contrastive training procedure intended to improve a model's ability to distinguish relevant from irrelevant long-range context during training, reporting improved effective context utilization relative to standard fine-tuning approaches.

5. Research Challenges and Limitations

Despite this body of work, several challenges remain unresolved. First, most benchmarks rely on synthetic or semi-synthetic retrieval tasks (e.g., needle-in-haystack, key-value lookup) that may not fully capture the demands of genuine research synthesis, which typically requires integrating, comparing, and reconciling claims across multiple documents rather than locating a single fact. Second, evaluation methodology varies considerably across studies, making cross-model and cross-paper comparisons difficult. Third, most published degradation curves are measured on English-language, text-only inputs; degradation dynamics for multilingual, multimodal, or structured (e.g., tabular) long-context inputs remain comparatively underexplored. Fourth, mitigation techniques such as positional interpolation and attention-sink preservation address specific proposed mechanisms of degradation but do not resolve the underlying question of whether current transformer training regimes systematically under-incentivize uniform long-range attention.

6. Research Gaps and Future Directions

We identify several directions warranting further research. First, there is a need for benchmarks that more closely approximate real research-synthesis workloads, such as multi-document literature review generation with ground-truth citation verification, rather than purely retrieval-oriented probes. Second, further work is needed to determine whether retrieval-augmented approaches (Lewis et al., 2020; Ram et al., 2023) fully substitute for large native context windows in synthesis tasks, or whether hybrid architectures combining retrieval with extended context offer complementary benefits. Third, the relationship between context-window degradation and citation or attribution errors in generated research syntheses — that is, whether degradation contributes to hallucinated or misattributed claims when source material is unevenly attended to — remains largely unstudied and is directly relevant to research-integrity concerns in AI-assisted literature review.

7. Conclusion

Context-window degradation represents a significant and empirically well-documented gap between the nominal context length advertised for large language models and their effective ability to integrate information uniformly across that length. For long-document research synthesis tasks in particular, this gap has direct practical consequences: material located in the middle of a large input corpus may be systematically under-weighted relative to material at the beginning or end, regardless of its scientific or argumentative importance. Mitigation strategies at the architectural, training, and system-design levels each offer partial remedies, but no single approach yet closes the gap fully. We conclude that research synthesis systems built on large language models should not treat nominal context length as a guarantee of uniform coverage, and should instead incorporate explicit retrieval, chunking, position-aware prompting, and verification steps to reduce the risk of uneven or biased synthesis.

References

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Chat Evidence: <https://claude.ai/share/ffa8a3e5-d6ae-4e02-bc68-a3c7bac53608>

PART D — Paper Profile and Reference Inventory

Measure	Value
Paper title	Context-Window Degradation and Its Impact on Long-Document Research Synthesis
Approximate pages	~7 pages (single-spaced equivalent)
Approximate word count	2,330 words
Number of sections	9 (Abstract, Intro, Background, Main Discussion, Current Approaches, Challenges, Gaps, Conclusion, References)
Total number of references	18
Approximate in-text citation occurrences	~26
Did AI warn you to verify references?	No explicit warning was given in the paper text itself

D1. Reference Inventory

Classified by best available identifier, priority: DOI → PMID → arXiv → URL only → none.

Reference Type	Count
DOI available	0
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	16
URL only	2
No persistent identifier supplied	0
TOTAL	18

Note: Ref #10 (Ram et al., TACL) was generated with a journal citation but no DOI supplied in the paper text, so it is classified under arXiv ID available (its arXiv preprint counterpart is well known) rather than DOI — this is exactly the kind of incomplete-identifier gap the lab asks students to notice.

PART E — Selected References for Audit

18 references > 10, so the sampling rule applies: first 3 + last 3 + 4 evenly spaced from the middle → bibliography numbers 1, 2, 3, 5, 8, 11, 14, 16, 17, 18.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	5
R5	8
R6	11
R7	14
R8	16
R9	17
R10	18

PART F — Plausibility Test (recorded BEFORE verification)

Ref.	Initial Impression	Plausibility (1–5)	Predict Genuine?
R1	Extremely well-known landmark paper; author names, venue, and arXiv ID all look exactly right.	5	Yes
R2	Recognizable, widely cited paper on this exact sub-topic; author list and framing look authentic.	5	Yes
R3	Plausible NVIDIA-style benchmark paper with a real-sounding acronym and full author list.	4	Yes

R4	Specific, technical title; author names sound like a real small team, not fabricated.	4	Yes
R5	Distinctive named method ("YaRN"); this kind of quirky acronym is typical of genuine papers.	4	Yes
R6	Well-known method name (ALiBi) with a memorable title; looks genuine.	5	Yes
R7	Plausible but less iconic benchmark name; could conceivably be a near-miss/franken-title.	3	Yes
R8	Corporate blog citation — less rigorous format, URL could easily be wrong or stale even if real.	3	Yes
R9	Informal GitHub source; very plausible given how well-known this project is in the field.	4	Yes
R10	Specific technical title ("Ring Attention"); sounds like a real systems paper.	4	Yes

Critical rule respected: these predictions were fixed before Part G verification began and were not edited afterward.

PART G — Citation Authenticity and Metadata Audit

Verification method: For each sampled reference, the exact arXiv ID / URL and author string given in the AI-generated paper were checked against arXiv.org, Semantic Scholar, Hugging Face Papers, GitHub, ACL Anthology, and (for the Anthropic blog) a direct search for the live post. No second AI chatbot's claim was used as evidence — only primary/scholarly sources.

G5. Verification Evidence

Ref.	Found?	Title	Authors	Year	Venue	ID Match	Source(s)	ID Checked	What Was Found
R1	Y	Y	Y	Y	Y	Y	arXiv.org (1706.03762)	arXiv	Exact match: title, all 8 authors, year 2017, NeurIPS venue.
R2	Y	Y	Y	Y	Y	Y	arXiv.org (2307.03172); Semantic Scholar	arXiv	Exact match. Note: later formally published in TACL (2024) — arXiv ID cited is still correct and resolves properly.
R3	Y	Y	N	Y	Y	Y	arXiv.org (2404.06654); NVIDIA/RULER GitHub	arXiv	Real NVIDIA paper, correct title/year/venue, but AI's author list omitted co-author Yang Zhang (8 authors, not 7).
R4	Y	Y	Y	Y	Y	Y	arXiv.org (2306.15595)	arXiv	Exact match: title, all 4 authors, year, abstract topic.
R5	Y	Y	Y	Y	Y	Y	arXiv.org (2309.00071)	arXiv	Exact match: title, all 4 authors, year.
R6	Y	Y	Y	Y	Y	Y	arXiv.org (2108.12409)	arXiv	Exact match: title, all 3 authors. Year cited as 2022 (v2 revision date) vs. 2021 (v1) — both used interchangeably in

									the literature, not a real discrepancy.
R7	Y	Y	Y	Y	Y	Y	arXiv.org (2305.14196); ACL Anthology	arXiv	Exact match: title, all 5 authors, year. Paper also later got an ACL DOI, which the AI did not include — an omission, not an error.
R8	Y	Y	N/A	Y	N	N	Web search for the post content; live Anthropic/Claude blog	URL	Content and authorship (Anthropic) are genuine and the post exists, but the AI-supplied URL (anthropic.com/news/...) does not match the post's current live location (claude.com/blog/claude-2-1-prompting).
R9	Y	Y	Y	Y	Y	Y	GitHub (gkamradt/LLMTest_NeedleInAHaystack)	URL	Exact match: repo exists, author, title, and URL all correct.
R10	Y	Y	Y	Y	Y	Y	arXiv.org (2310.01889); Semantic Scholar	arXiv	Exact match: title, all 3 authors, year.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Fully verified against arXiv — exact match on all fields.
R2	A	Fully verified against arXiv — exact match on all fields.
R3	B	Real, correct paper, but AI's author list dropped one of the 8 real co-authors.
R4	A	Fully verified against arXiv — exact match on all fields.
R5	A	Fully verified against arXiv — exact match on all fields.
R6	A	Fully verified against arXiv — exact match on all fields.
R7	A	Fully verified against arXiv — exact match on all fields.
R8	B	Real Anthropic blog post with correct authorship/content, but AI cited a stale/incorrect URL.
R9	A	Fully verified — GitHub repo and URL match exactly.
R10	A	Fully verified against arXiv — exact match on all fields.

PART G — Scoring

Count of final classifications: A = 8, B = 2, C = 0, D = 0, E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E = 4(8) + 3(2) + 1(0) + 1(0) = 32 + 6 = 38

Authenticity Score = (38 / (4 × 10)) × 100 = (38 / 40) × 100 = 95.0 / 100

Interpretation: 90–100 → Excellent authenticity.

G7. Prediction Accuracy

Part F predicted 'Genuine = Yes' for all 10 references. Treating a code of A or B as 'the publication genuinely exists' (correct on the existence question) and C/D/E as 'not genuine', all 10 predictions were correct on the existence question — though two of those ten (R3, R8) were only correctly predicted 'exists' while missing that their metadata had a flaw, which Part F was not designed to catch.

Correct predictions = 10 Total predictions = 10 Prediction Accuracy = 100%

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	18
References deeply audited	10
A: Verified	8
B: Wrong metadata	2
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	95.0 / 100
Prediction Accuracy	100%

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Mostly yes in this run — but this is precisely the risk the lab is designed to expose. Because the topic (long-context LLM research) sits inside the AI's dense, well-documented training data, most generated citations happened to be real, high-profile papers the model had 'seen' many times. The two flaws that did appear — an incomplete author list and a stale URL — were both invisible from formatting alone; they only surfaced once the underlying identifier was actually checked. A confident, well-formatted citation was not, on its own, proof of correctness.

2. Did all genuine references actually support the claims for which they were cited?

This worksheet's Part G scope covered existence and metadata accuracy, not full claim-by-claim entailment (that is Part H's job). Spot-checking during verification suggested the citations were used in ways consistent with each paper's actual findings (e.g., Liu et al. was cited specifically for the U-shaped position-bias result, which matches the real abstract). A dedicated Part H pass would be needed to confirm this systematically across the whole paper rather than the 10-reference sample.

3. What was the most serious citation-integrity problem observed?

The most serious issue was not outright fabrication but silent metadata drift: an author was dropped from a real citation (R3), and a real source was pointed to by an outdated URL (R8). These errors are more dangerous than obvious fabrications precisely because they are easy to overlook — the paper, title, and general claim all check out at a glance, so a reader has little reason to dig further unless they deliberately verify every field.

4. What is the single most important lesson learned from this lab?

A citation's surface plausibility (correct-looking author names, a real-sounding venue, a properly formatted arXiv ID) is not the same thing as verification. Even when an AI model is drawing on genuinely well-known, real literature, it can still quietly drop an author or reference a superseded link —

errors invisible without checking each field against a primary source. Systematic, field-by-field verification — not confidence in how 'clean' a reference list looks — is the only reliable defense against citation integrity failures in AI-assisted research.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name (e.g., Lab1_IEC2026001_RahulSharma)

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Field	Value
Student Name	Harsh Kumar Singh
Roll Number	PSE2026002
Signature	Harsh
Date	21/08/2026